

2021

YEAR 11
YEARLY
EXAMINATION

Mathematics Standard 2

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- Write your Student ID below and on page 6
- NESA approved calculators may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks:
80

Section I — 10 marks (pages 1–5)

- Attempt Questions 1–10
- Allow about 20 minutes for this section

Section II — 70 marks (pages 6–21)

- Attempt Questions 11–27
- Allow about 1 hour 40 minutes for this section

STUDENT ID: _____



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This paper is used with the understanding that it has a Security Period. ©Total Education Centre

Section I**10 marks****Attempt Questions 1–10****Allow about 20 minutes for this section**

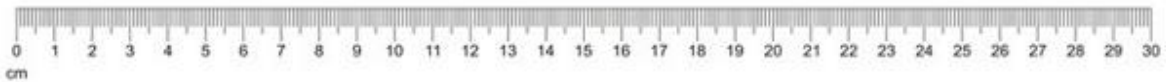
Use the multiple-choice answer sheet for Questions 1–10.

-
- 1** Daniel buys a new computer for \$988 including 10% GST.

What is the price of the computer before GST?

- A. \$889.20
- B. \$898.18
- C. \$988.00
- D. \$1086.80

- 2** A 30 cm ruler has increments marked in cm and mm. A measurement of 22.5 cm is taken using this ruler.



What is the percentage error of this measurement?

- A. 0.22 %
 - B. 0.5 %
 - C. 2.22 %
 - D. 2.25 %
- 3** A toy car manufacturer checks every fifth car on a production line to ensure product quality.

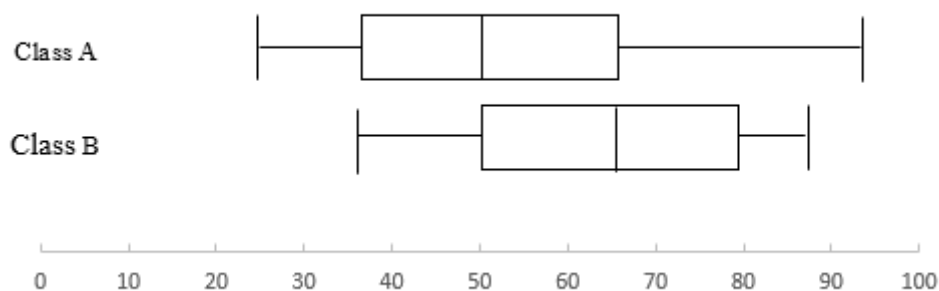
What method of sampling is being used?

- A. Random
- B. Stratified
- C. Systematic
- D. Self-selected

- 4 An experiment has three distinct outcomes. α , β , and γ . The probability of α occurring is 0.43 and β is four times more likely to occur than γ .

What is the probability of β occurring?

- A. 0.11
 - B. 0.14
 - C. 0.46
 - D. 0.57
- 5 The chart below shows the scores of two classes on a science test.



Which is most accurate about the results?

- A. Class B has a larger range than Class A
 - B. The interquartile ranges of the two classes are the same
 - C. The median of Class B is equal to the first quartile of Class A
 - D. The median of Class A is equal to the third quartile of Class B
- 6 The average off-peak rate for domestic electricity consumption is \$0.18 per kWh and the average peak rate is \$0.26 per kWh.

What is the cost of running a 160-watt television for two hours during peak time, every day for an entire year?

- A. \$21.02
- B. \$30.37
- C. \$57.60
- D. \$83.20

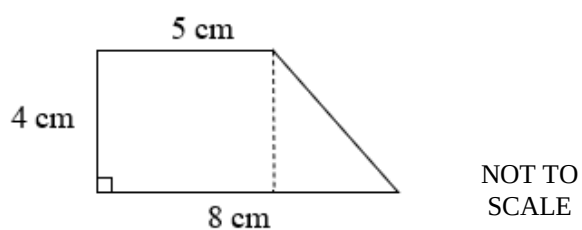
- 7 In 2020 there were 4 006 974 students enrolled in Australian schools.

How would this number be expressed in standard form correct to two significant figures?

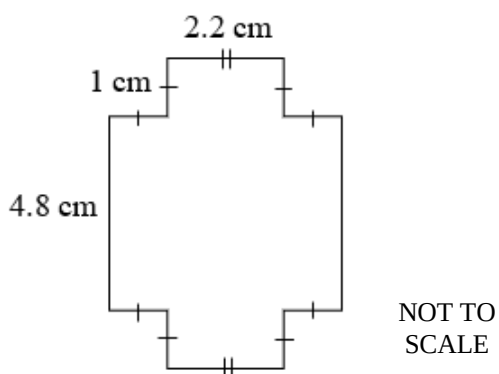
- A. 4.0×10^6
- B. 4.006×10^6
- C. 40×10^5
- D. 40.06×10^5

- 8 Which shape has the longest perimeter?

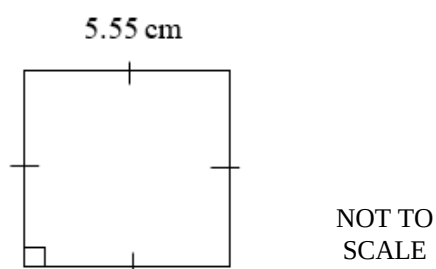
A.



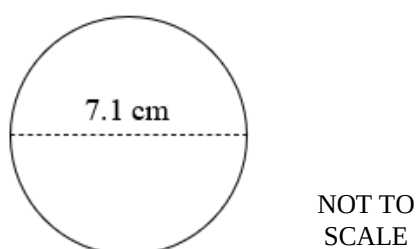
B.



C.



D.



- 9** Gloria is ill and takes medicine. The adult dosage of the medicine is 12 mL.

Gloria's one-and-a-half year old daughter also becomes ill and requires medicine.

Using Fried's formula, determine the correct dose of the medicine Gloria should give to her daughter.

Fried's formula: $\text{Dosage (for Children 1 – 2 years)} = \frac{\text{age (in months)} \times \text{adult dosage}}{150}$

- A. 0.12 mL
 - B. 0.2 mL
 - C. 1.2 mL
 - D. 1.44 mL
- 10** Garry bought a car at the beginning of 2013 for \$28 990. The car depreciates at a rate of 5% per annum straight-line depreciation.

What is the salvage value of the car at the start of 2021?

- A. \$10 146.50
- B. \$11 596
- C. \$17 394
- D. \$18 843.50

End of Section I

2021

**YEAR 11 YEARLY
EXAMINATION**

Student ID: _____

Mathematics Standard 2 Section II Answer Booklet

70 marks

Attempt Questions 11–27

Allow about 1 hour 40 minutes for this part

Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Your responses should include relevant mathematical reasoning and/or calculations
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
- Write your Student ID above

Please turn over

Question 11 (2 marks)

A construction worker measures a 2.413 metre piece of timber using a tape measure with millimetre markings. **2**

Calculate the upper and lower limit of accuracy. Give your answer in millimetres.

Question 12 (2 marks)

A shirt costs \$84 before being discounted by 15%. The store is having a 30% off sale and takes this amount from the discounted cost. **2**

What is the final sale price of the shirt?

Question 13 (3 marks)

A flight from Rome, Italy (GMT +2) to Chicago, USA (GMT -5) takes 14 hours 50 minutes. **3**

If the flight leaves Rome at 6pm Saturday local time, what time would it be in Chicago when the plane lands?

Question 14 (2 marks)

Lucy rents a car from a hire company. The cost to hire the car is \$15 per hour plus a hire fee of \$100. Lucy pays \$355.

2

By developing a linear equation, determine how many hours she hired the car for.

Question 15 (4 marks)

Jordan weighs 63 kg and attends a Christmas party.

4

He has his first alcoholic drink at 4 pm and continues drinking until 10 pm when his blood alcohol content is 0.035.

Using the BAC formula, calculate how many standard drinks Jordan has consumed.

$$BAC_{male} = \frac{10N - 7.5H}{6.8M}$$

where N is the number of standard drinks consumed, H is the number of hours of drinking, and M is the person's weight in kilograms.

Question 16 (3 marks)

Travis is catching a plane from International Airport. He needs to do this in two steps.

3

First, he will catch a bus from Saywell Road to Glenfield Station. Secondly, from Glenfield Station he will catch a train to International Airport.

These are the bus and train timetables he would use.

Step 1 - Timetable: to Glenfield Station:

Ingleburn Station	-	-	05:59	06:22	06:53	07:16	07:56
Macquarie Fields High School, School Grounds, Macquarie Fields	-	-	-	-	-	-	-
Saywell Rd opp Pritchard Rd, Macquarie Fields	-	05:28	06:05	06:29	07:00	07:23	08:03
Evelyn St near Eucalyptus Dr, Macquarie Fields	-	05:32	06:11	06:35	07:06	07:29	08:09
Victoria Rd opp Macquarie Fields TAFE Campus, Macquarie Fields	-	05:36	06:17	06:41	07:12	07:36	08:15
Interline Bus Depot Atchison Rd, Macquarie Fields	-	05:38	06:19	06:43	07:15	07:39	08:18
Glenfield Station, Railway Pde, Glenfield	05:24	05:41	06:23	06:47	07:18	07:43	08:21
Glenfield Rd near Campbelltown Rd, Glenfield	05:30	05:47	06:30	06:54	07:26	07:53	08:29
Hume Hwy after Graham Ave, Casula	05:35	05:52	06:35	06:59	07:31	08:01	08:34
Liverpool Interchange, Liverpool	05:43	06:01	06:45	07:11	07:44	08:17	08:53

Step 2 - Timetable: Glenfield Station to airports:

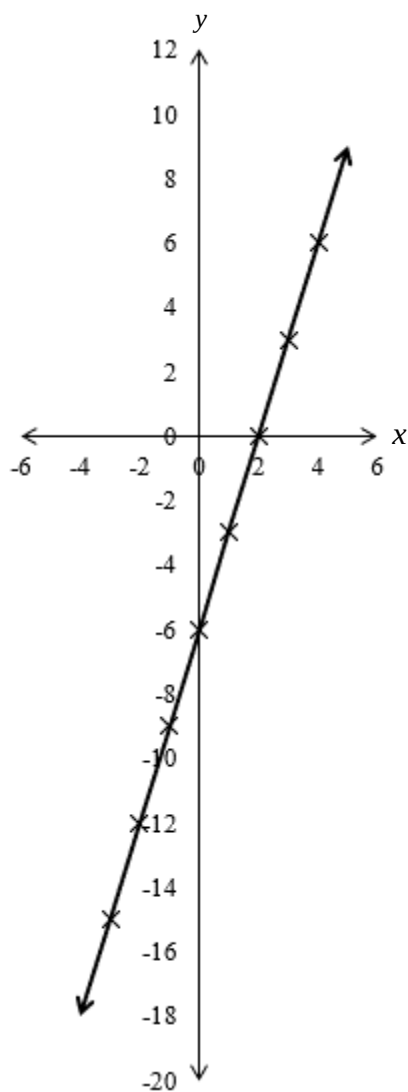
Glenfield	ARR	07:07	-	07:14	07:22	-	07:29	07:34	07:37	-
Glenfield	DEP	07:08	-	07:15	07:23	-	07:30	07:35	07:38	-
Holsworthy		07:13	-	07:20	07:28	-	07:35	07:40	07:43	-
East Hills		07:17	-	-	07:32	-	-	-	07:47	-
Panania		07:19	-	-	07:34	-	-	-	07:49	-
Revesby		07:22	07:13	-	07:37	07:28	-	07:46	07:52	07:43
Padstow		-	07:16	07:28	-	07:31	07:43	-	-	07:46
Riverwood		-	07:18	07:30	-	07:33	07:45	-	-	07:48
Narwee		-	07:21	-	-	07:36	-	-	-	07:51
Beverly Hills		-	07:23	-	-	07:38	-	-	-	07:53
Kingsgrove		-	07:26	-	-	07:41	-	-	-	07:56
Bexley North		-	07:28	-	-	07:43	-	-	-	07:58
Bardwell Park		-	07:30	-	-	07:45	-	-	-	08:00
Turrella		-	07:33	-	-	07:48	-	-	-	08:03
Wolli Creek		-	07:35	07:41	-	07:50	07:56	07:59	-	08:05
International Airport		-	07:37	07:43	-	07:52	07:58	08:01	-	08:07
Domestic Airport		-	07:39	07:45	-	07:54	08:00	08:03	-	08:09

What is the latest time Travis can catch the bus from Saywell Rd to arrive at the airport before 8 am?

Question 17 (4 marks)

- (a) Create an equation for the following straight line.

2



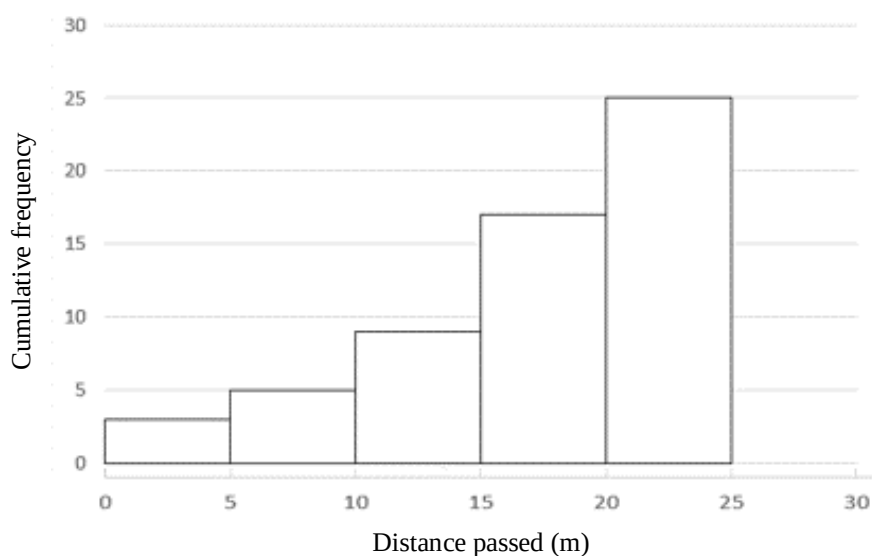
- (b) For the line above, calculate x if $y = -21$.

2

Question 18 (4 marks)

A rugby team recorded how far players could pass a ball, in metres.

The graph below shows the distance the players could pass and the cumulative frequency.



- (a) Three players passed between 25 and 30 metres. Display this on the graph. **1**
- (b) Draw the cumulative frequency histogram of this information on the graph. **1**
- (c) What percentage of players passed between 5 and 20 metres? **2**

Question 19 (3 marks)

The table shows the average energy used, in kilojoules per kilogram of body mass, by a person rowing a canoe for 10 minutes at different speeds. **3**

<i>Rowing speed</i>	<i>Energy used in 10 minutes</i>
12 km/h	2.06 kJ/kg
18 km/h	3.15 kJ/kg

Cassandra, who weighs 58 kg, and Molly, who weighs 63 kg both go canoeing.

Cassandra rows at 18 km/h for 40 minutes and Molly rows at 12 km/h for one hour.

Who burns off more kilocalories during their row, and by how many? Give your answer to the nearest kilocalorie.

(1 kilocalorie = 4.184 kJ).

Question 20 (3 marks)

Charlotte invests \$3100 into a savings account with no further deposits. She wants to save for a holiday that will cost \$4250 in four years' time. **3**

At what rate of simple interest must she invest the \$3100 to be able to afford her holiday? Give your answer to two significant figures.

Question 21 (5 marks)

Eliza buys a new car for \$35 190 and is required to pay stamp duty. Stamp duty for vehicles in NSW is calculated at \$3 per \$100, or part thereof, up to \$45 000 then an additional \$5 per \$100, or part thereof, for the value over \$45 000.

- (a) Calculate the stamp duty Eliza must pay on her new car.

2

- (b) Eliza takes out compulsory third-party insurance on the car costing \$143.50 per month and pays registration of \$82 every quarter. She also must have the car serviced every 6 months, costing \$200 each service. In addition, Eliza spends an average of \$54.20 a week on fuel.

3

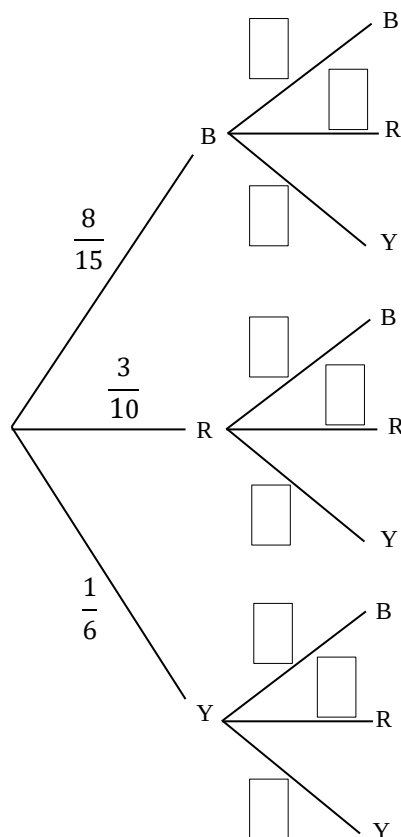
Calculate the total cost of purchasing and running the car for the first year.

Question 22 (5 marks)

In a fishpond there are thirty fish. Sixteen of the fish are blue, nine are red and five are yellow.

Jack is cleaning the fishpond and needs to remove the fish. He catches two fish without returning them.

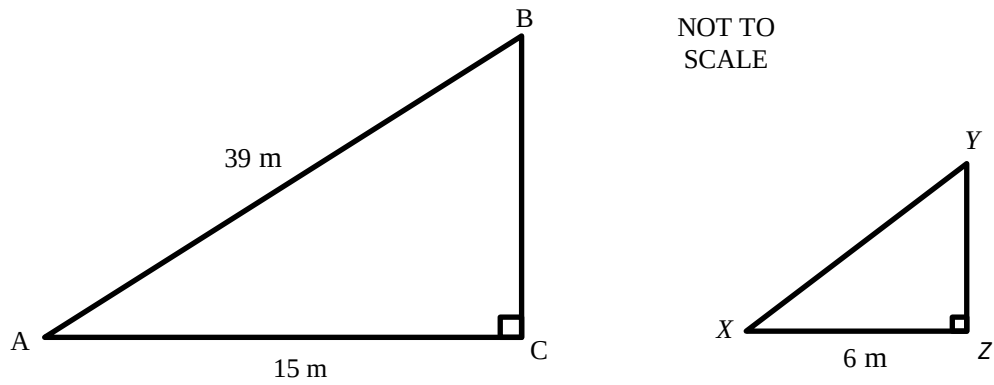
- (a) Complete the probability tree on the diagram. The first branch is simplified for you. 3



- (b) Calculate the probability of catching two different coloured fish to two decimal places. 2

Question 23 (4 marks)

Triangles ABC and XYZ are similar.

4

Calculate the area of triangle XYZ.

Question 24 (5 marks)

Peter works as a construction supervisor and is calculating his tax return.

5

This is an extract from Peter's payment summary.

PAYG payment summary – individual non-business
Payment summary for year ending 30 June

Payee details
Peter Short
19 Pickle Lane
Realtown, NSW, 2400

NOTICE TO PAYEE
If this payment summary shows an amount in the total tax withheld box you must lodge a tax return. If no tax was withheld you may still have to lodge a tax return.
For more information on whether you have to lodge, or about this payment and how it is taxed, you can:
■ refer to *TaxPack*
■ phone 13 28 61 between 8.00am and 6.00pm (EST), Monday to Friday, or
■ visit www.ato.gov.au

Period during which payments were made 01/07/2020 to 30/06/2021

Payee's tax file number 123 456 789

TOTAL TAX WITHHELD \$ 52 596.00

		Lump sum payments
Gross Income	\$ 167 920.00	-
Other Income	\$ 14 795.00	-
Deductions	\$ 832.00	-
Taxable Income	\$ XXXXXXXX	-

The table shows tax payable in Australia according to taxable income.

<i>Taxable income (\$)</i>	<i>Tax payable</i>
\$0 – \$18 200	Nil
\$18 201 – \$45 000	19 cents for each \$1 over \$18 200
\$45 001 – \$120 000	\$5 092 plus 32.5 cents for each \$1 over \$45 000
\$120 001 – \$180 000	\$29 467 plus 37 cents for each \$1 over \$120 000
\$180 001 and over	\$51 667 plus 45 cents for each \$1 over \$180 000

The Medicare levy in Australia is calculated at 2% of taxable income.

Determine if Peter will receive a refund or owe money, and how much. Justify your answer with calculations.

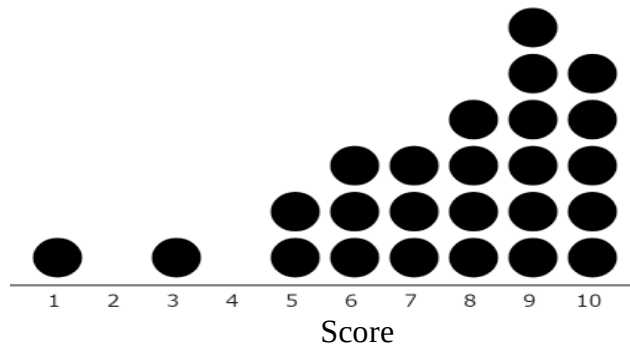
lines for Question 24 continue on page 17

Question 24 (continued)

End of Question 24

Question 25 (6 marks)

This dot plot shows the scores achieved on a spelling test by a class of Year 6 students.



(a) Describe the skewness of the data. 1

(b) Determine if there are any outliers in the data. Justify your answer with calculations. 2

(c) A student who was absent on the day of the original test achieves a score of 6. 3
How does this score impact upon the mean, median and mode of the class data?

Question 26 (7 marks)

- (a) A carpenter earns \$2010.20 a fortnight when he works 38 hours a week. 2

Calculate his equivalent hourly wage.

- (b) Annual leave loading is calculated at 17.5% of four weeks wages. 3

Calculate the carpenter's total income for the four weeks leave period, assuming he averages \$2010.20 per fortnight.

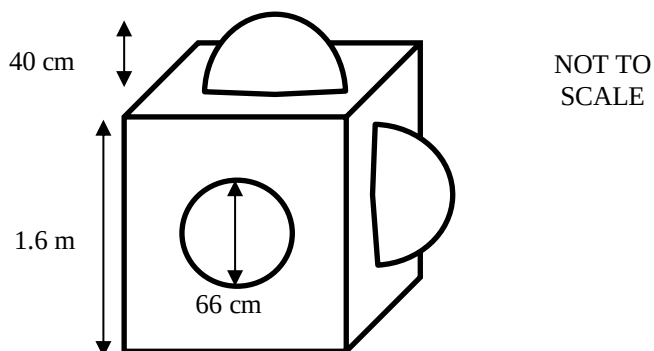
- (c) The same carpenter works a 40-hour week at his base rate and works an additional eight hours of overtime at time and a half. 2

Calculate his income for that week.

Question 27 (8 marks)

A playground is to be constructed using a series of connected modules.

Below is a diagram of one of the modules to be constructed.



This module is made of an enclosed cube with side lengths of 1.6 metres. On two sides of the cube are identical hemispheres with heights of 40 cm. On the front of the cube there is a circular hole for entry with a diameter of 66 cm.

- (a) Calculate the surface area of the playground module correct to the nearest square metre.

5

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Question 27 continues on page 21

Question 27 (continued)

- (b) Calculate the volume of the playground module correct to two decimal places. **3**

End of paper

REFERENCE SHEET

Measurement**Limits of accuracy**

Absolute error = $\frac{1}{2} \times \text{precision}$

Upper bound = measurement + absolute error

Lower bound = measurement – absolute error

Length

$$l = \frac{\theta}{360} 2\pi r$$

Area

$$A = \frac{\theta}{360} \pi r^2$$

$$A = \frac{h}{2} (a + b)$$

$$A \approx \frac{h}{2} (d_f + d_l)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3} Ah$$

$$V = \frac{4}{3} \pi r^3$$

Trigonometry

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \cos A = \frac{\text{adj}}{\text{hyp}}, \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2} ab \sin C$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Financial Mathematics

$$FV = PV (1 + r)^n$$

Straight-line method of depreciation

$$S = V_o - Dn$$

Declining-balance method of depreciation

$$S = V_o (1 - r)^n$$

Statistical Analysis

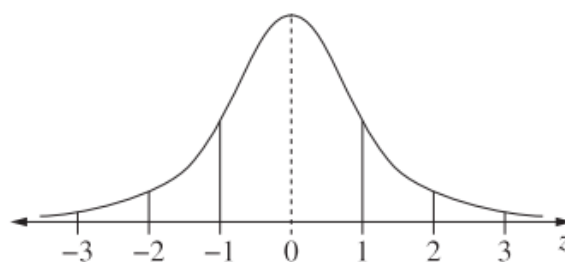
An outlier is a score

less than $Q_1 - 1.5 \times IQR$

or

more than $Q_3 + 1.5 \times IQR$

$$z = \frac{x - \bar{x}}{s}$$

Normal distribution

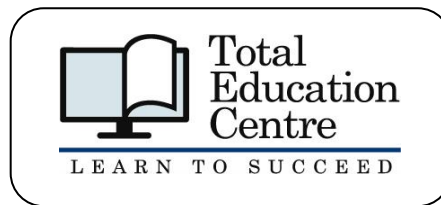
- approximately 68% of scores have z-scores between -1 and 1
- approximately 95% of score have z-scores between -2 and 2
- approximately 99.7% of score have z-scores between -3 and 3

STUDENT ID _____

2021 Mathematics Standard 2 Year 11 Examination**Section I Answer Sheet****10 marks****Attempt Questions 1–10****Allow about 20 minutes for this section**

Select the alternative A, B, C, or D that best answers the question. Fill in the response circle completely.

- | | | | | |
|-----------|-----|-----|-----|-----|
| 1 | A ○ | B ○ | C ○ | D ○ |
| 2 | A ○ | B ○ | C ○ | D ○ |
| 3 | A ○ | B ○ | C ○ | D ○ |
| 4 | A ○ | B ○ | C ○ | D ○ |
| 5 | A ○ | B ○ | C ○ | D ○ |
| 6 | A ○ | B ○ | C ○ | D ○ |
| 7 | A ○ | B ○ | C ○ | D ○ |
| 8 | A ○ | B ○ | C ○ | D ○ |
| 9 | A ○ | B ○ | C ○ | D ○ |
| 10 | A ○ | B ○ | C ○ | D ○ |



2021 Mathematics Standard 2 Year 11 Marking Guidelines

Section I

Multiple-choice Answer Key

Question	Answer
1	B
2	A
3	C
4	C
5	B
6	B
7	A
8	D
9	D
10	C

Section II

Note: An incorrect answer in a previous part will not necessarily preclude students from achieving full marks in a later part. Answers here are based on correct prior part answers. Marking will need to adapt to pursue correct method with the use of incorrect prior parts.

Question 11 (2 marks)

Criteria	Marks
• Provides correct answer	2
• Makes progress towards answer	1

Sample answer:

Absolute error = ± 0.5 mm

$$2.413 \text{ m} \times 100 \times 10 = 2413 \text{ mm}$$

Upper limit = Measurement + absolute error

$$= 2413 + 0.5$$

$$= 2413.5 \text{ mm}$$

Lower limit = Measurement – absolute error

$$= 2413 - 0.5$$

$$= 2412.5 \text{ mm}$$

Question 12 (2 marks)

Criteria	Marks
• Provides correct answer	2
• Makes progress towards answer	1

Sample answer:

$$0.85 \times 84$$

$$= \$71.40$$

$$0.7 \times 71.4$$

$$= \$49.98$$

Question 13 (3 marks)

Criteria	Marks
• Provides correct answer	3
• Makes progress towards answer	2
• Provides some relevant information	1

Sample answer:

Time the flight lands (Rome time) = 6 pm + 14 h 50 min
= 8:50 am Sunday

Time difference = 2 + 5 = 7 hours (Chicago behind)

Time the flight lands (Chicago time) = 8:50 am – 7 h
= 1:50 am Sunday

Question 14 (2 marks)

Criteria	Marks
• Provides correct answer	2
• Makes progress towards answer	1

Sample answer:

$$C = 15h + 100$$

$$355 = 15h + 100$$

$$255 = 15h$$

$$h = \frac{255}{15}$$

$$h = 17 \text{ hours}$$

Question 15 (4 marks)

Criteria	Marks
• Provides correct answer	4
• Makes significant progress towards answer	3
• Makes progress towards answer	2
• Provides some relevant information	1

Sample answer:

Rearrange formula:

$$N = \frac{BAC \times 6.8M + 7.5H}{10}$$

$$H = 10 - 4 = 6 \text{ hours}$$

$$N = \frac{0.035 \times (6.8 \times 63) + (7.5 \times 6)}{10}$$

$$N = 5.999$$

$$N = 6 \text{ drinks}$$

Question 16 (3 marks)

Criteria	Marks
• Provides correct answer	3
• Makes progress towards answer	2
• Provides some relevant information	1

Sample answer:

Latest Train arrives at the airport at 7:58 am

To catch this, he must arrive at Glenfield Station before 7:30 am.

Latest arrival at Glenfield Station before 7:30 is 7:18.

To do this he must catch the 7:00 bus from Saywell Rd.

Latest bus is the 7:00 am.

Question 17 (4 marks)

a

Criteria	Marks
• Provides correct answer	2
• Makes progress towards answer	1

Sample answer:

$$Y = mx + b$$

$$m = \text{gradient} = \frac{\text{rise}}{\text{run}}$$

$$= \frac{3}{1} = 3$$

$$b = Y\text{-intercept} = -6$$

$$y = 3x - 6$$

b

Criteria	Marks
• Provides correct answer	2
• Makes progress towards answer	1

Sample answer:

$$Y = -21$$

$$Y = 3x - 6$$

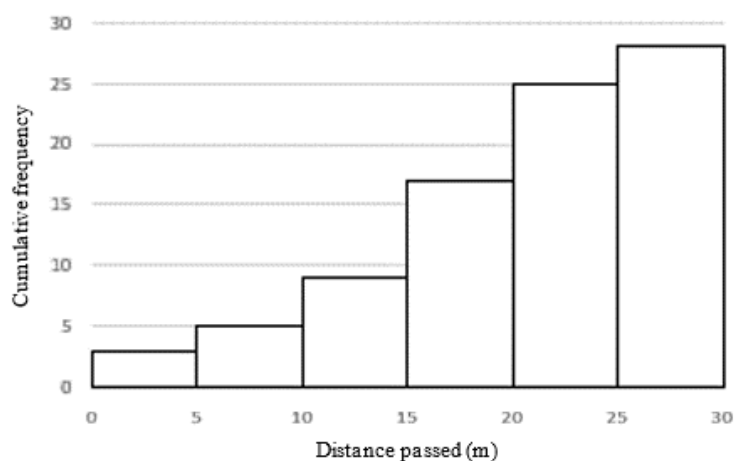
$$-21 = 3x - 6$$

$$-15 = 3x$$

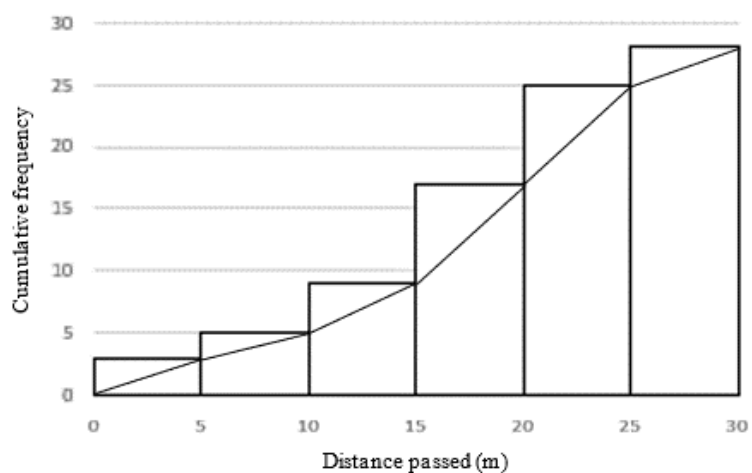
$$X = -5$$

Question 18 (4 marks)**a**

	Marks
Provides correct answer	1

Sample answer:**b**

	Marks
Draws correct histogram	1

Sample answer:**c**

Criteria	Marks
Provides correct answer	2
Makes progress towards answer	1

Sample answer:

$$17 - 3 = 14$$

$$\text{Total} = 28$$

$$\frac{14}{28} \times 100 =$$

$$50\%$$

Question 19 (3 marks)

Criteria	Marks
• Provides correct answer	3
• Makes progress towards answer	2
• Provides some relevant information	1

Sample answer:

Cassandra burns $58 \times 3.15 \times 4 = 730.8$ kJ

Molly burns $63 \times 2.06 \times 6 = 778.68$ kJ

Molly burns 47.88 kJ more

$47.88 \div 4.184 = 11.44$ kilocalories

$\therefore$ Molly burns 11 kilocalories more than Cassandra during their rows.

Question 20 (3 marks)

Criteria	Marks
• Provides correct answer	3
• Makes progress towards answer	2
• Provides some relevant information	1

Sample answer:

$I = Prn$

Interest = $4250 - 3100 = \$1150$

$1150 = 4250 \times r \times 4$

$r = \frac{1150}{17000}$

$= 0.0676$

$= 6.8\%$ p.a.

Question 21 (5 marks)

a

Criteria	Marks
• Provides correct answer	2
• Makes progress towards answer	1

Sample answer:

$0.03 \times 35\,200 =$

$= \$1056$

b

Criteria	Marks
• Provides correct answer	3
• Makes progress towards answer	2
• Provides some relevant information	1

Sample answer:Cost of CTP = $143.50 \times 12 = \$1722$ Cost of rego = $82 \times 4 = \$328$ Services = $200 \times 2 = \$400$ Fuel = $54.20 \times 52 = \$2818.40$

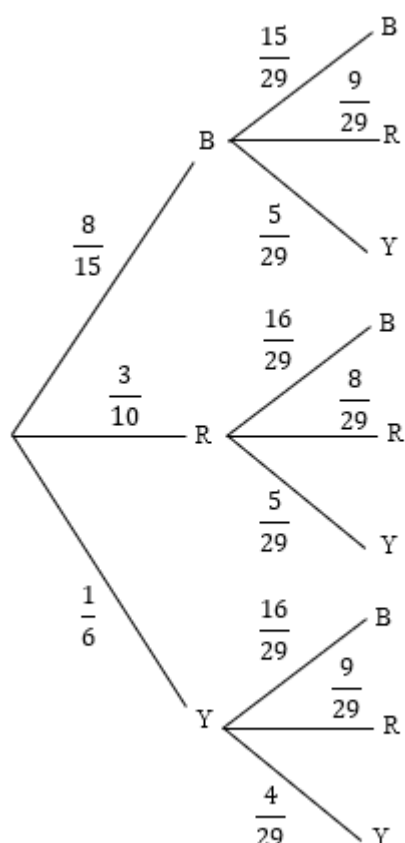
$$35\,190_{\text{car cost}} + 1056_{\text{Stamp duty}} + 1722_{\text{CTP}} + 328_{\text{Rego}} + 400_{\text{service}} + 2818.40_{\text{fuel}}$$

$$= \$41\,514.40$$

Question 22 (5 marks)

a

Criteria	Marks
• Provides correct answer	3
• Makes progress towards answer	2
• Provides some relevant information	1

Sample answer:

Question 22 (continued)**b**

Criteria	Marks
• Provides correct answer	2
• Makes progress towards answer	1

Sample answer:

$$\rho = 1 - \left(\left(\frac{8}{15} \times \frac{15}{29} \right)_{\rho-BB} + \left(\frac{3}{10} \times \frac{8}{29} \right)_{\rho-RR} + \left(\frac{1}{6} \times \frac{4}{29} \right)_{\rho-YY} \right)$$

$$\rho = 0.6184$$

$$\rho = 0.62$$

Question 23 (4 marks)

Criteria	Marks
• Provides correct answer	4
• Makes significant progress towards answer	3
• Makes progress towards answer	2
• Provides some relevant information	1

Sample answer:

$$a^2 + b^2 = c^2$$

$$a^2 = 39^2 - 15^2$$

$$a = \sqrt{1296}$$

$$a = 36$$

$$x = \frac{6}{15} \times 36$$

or

$$x = \frac{36}{15} \times 6$$

$$x = 14.4 \text{ m}$$

$$\text{Area} = \frac{1}{2} \times 6 \times 14.4$$

$$\text{Area} = 43.2 \text{ m}^2$$

Question 24 (5 marks)

Criteria	Marks
• Provides correct answer, with clear working and steps indicated	5
• Provides answer with one minor error/omission	4
• Makes significant progress towards answer	3
• Makes progress towards answer	2
• Provides some relevant information	1

Sample answer:

Taxable income = $167\,920 + 14\,795 - 832$
 $= \$181\,883$

Tax payable = $51\,667 + 0.45 \times (183\,547 - 180\,000)$
 $= 51\,667 + 847.35$
 $= \$52\,514.35$

Medicare Levy = $0.02 \times 181\,883$
 $= \$3\,637.66$

Tax payable
 $= \$56\,152.01$

Amount owed = $56\,152.01 - 52\,596$
 $= \$3\,556.01$

□ Peter owes \$3556.01 in tax.

Question 25 (6 marks)

a

	Marks
• Provides correct answer	1

Sample answer:

Negatively skewed

b

Criteria	Marks
• Provides correct answer	2
• Makes progress towards answer	1

Sample answer:

Outliers

Less than $Q1 - 1.5 \times IQR$ Or more than $Q3 + 1.5 \times IQR$

Median = 8

 $Q1 = 6$ $Q3 = 9$ $IQR = 9 - 6 = 3$ $Q3 + 1.5 \times IQR$ $9 + 1.5 \times 3 = 13.5$ $Q1 - 1.5 \times IQR$ $6 - 1.5 \times 3$ $= 1.5$ ☐ 1 is an outlier.**c**

Criteria	Marks
• Describes all three changes	3
• Describes two changes	2
• Describes one change	1

Sample answer:

Original mean = 7.56

New mean = 7.5

☐ Mean is decreased.

Original median = 8

New median = 8

☐ Median is unchanged.

Original mode = 9

New mode = 9

☐ Mode is unchanged.

Question 26 (7 marks)**a**

Criteria	Marks
• Provides correct answer	2
• Makes progress towards answer	1

Sample answer:

$$38 \times 2 = 76 \text{ hours per fortnight}$$

$$2010.20 \div 76 = \$26.45$$

b

Criteria	Marks
• Provides correct answer	3
• Makes progress towards answer	2
• Provides some relevant information	1

Sample answer:

$$2010.20 \times 2 = 4\,020.40$$

$$\text{Loading} = 0.175 \times 4\,020.40 \\ = 699.55$$

$$\text{Income} = 4\,020.40 + 699.55 \\ = \$4719.95$$

c

Criteria	Marks
• Provides correct answer	2
• Makes progress towards answer	1

Sample answer:

$$\$26.45 \times 40 = \$1058$$

$$\$26.45 \times 8 \times 1.5 = \$317.40$$

$$\text{Total pay} = 1058 + 317.40 \\ = \$1375.40$$

Question 27 (8 marks)**a**

Criteria	Marks
• Provides correct answer, with clear working and steps indicated	5
• Provides answer with one minor error/omission	4
• Makes significant progress towards answer	3
• Makes progress towards answer	2
• Provides some relevant information	1

Sample answer:Surface area of Cube= A_1

$$A_1 = 1.6 \times 1.6 \times 6$$

$$= 15.36 \text{ m}^2$$

Surface area of Hemispheres= A_2

$$A_2 = 4 \pi r^2$$

$$= 4 \times \pi \times 0.4^2$$

$$= 2.01 \text{ m}^2$$

Area of hemisphere base circles= A_3

$$A_3 = 2 (\pi r^2)$$

$$= 2 \times \pi \times 0.4^2$$

$$= 1.01 \text{ m}^2$$

Area of circular opening= A_4

$$A_4 = \pi r^2$$

$$= \pi \times 0.33^2$$

$$= 0.34 \text{ m}^2$$

Total area = $A_t = A_1 + (A_2 - A_3) - A_4$

$$A_t = 16.02 \text{ m}^2$$

$$16 \text{ m}^2$$

b

Criteria	Marks
• Provides correct answer	3
• Makes progress towards answer	2
• Provides some relevant information	1

Sample answer:Volume of cube= V_1

$$V_1 = 1.6 \times 1.6 \times 1.6$$

$$= 4.096 \text{ m}^3$$

Volume of hemispheres= V_2

$$V_2 = \frac{4}{3} \pi r^3$$

$$= \frac{4}{3} \times \pi \times 0.4^3$$

$$= 0.2680 \text{ m}^3$$

Total Volume= $V_t = V_1 + V_2$

$$= 4.36 \text{ m}^3$$

2021 Mathematics Standard 2 Year 11 Mapping Grid

Section I

Question	Marks	Syllabus outcomes
1	1	F1.1
2	1	M1.1
3	1	S1.1
4	1	S2
5	1	S1.2
6	1	M1.3
7	1	M1.1
8	1	M1.2
9	1	A1
10	1	F1.1

Section II

Question	Marks	Syllabus outcomes
11	2	M1.1
12	2	F1.1
13	3	M2
14	2	A1
15	4	A1
16	3	M2
17	4	A2
18	4	S1.1
19	3	M1.3
20	3	F1.1/A1
21	5	F1.3
22	5	S2
23	4	M1.2/A1
24	5	F1.2
25	6	S1.2
26	7	F1.2
27	8	M1.2